THE GYSIN CLASS OF A LOCAL COMPLETE INTERSECTION MORPHISM IS INDEPENDENT OF THE CHOSEN FACTORIZATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FF2 (sources/stacks-chow/pinned-original-source.tex line 13485).

Allowed standard prerequisites (editorial).

- Composition of Gysin classes and the Gysin class of a regular immersion. Same chapter, Lemmas labelled lemma-composition-regular-immersion and lemma-construction-gysin, cited at proof lines 13517-13523. Splits (i')^! = s^! o j^! and identifies j^! as a restriction of i^!.
- Regular immersions obtained from a section of a smooth morphism and from flat base change. Divisors, Lemma labelled divisors-lemma-section-smooth-regular-immersion and Lemma labelled divisors-lemma-flat-base-change-regular-immersion, cited at proof lines 13513-13517. Makes s and j regular immersions so that their Gysin classes exist.
- Composition of flat pullbacks and commutation of bivariant classes with flat pullback. Same chapter, Lemma labelled lemma-compose-flat-pullback, cited at proof lines 13505-13507, and the flat pullback commutation used at proof lines 13523-13525. Splits (g')^* and moves the flat pullback past j^!.
- The Gysin class of a section of a smooth morphism is the identity. Same chapter, Lemma labelled lemma-section-smooth, cited at proof lines 13525-13527. Finishes the reduction with s^! o p-bar^* = id.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Same-chapter pointer into the author environment situation at source line 1372; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Date: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02RD: sources/stacks-chow/pinned-original-source.tex, source lines 2287–2287. Same-chapter pointer into the author environment lemma at source line 2287; invoked at proof line 13506 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 067R: sources/stacks-chow/tags.txt, source lines 7980–7980. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 13516 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 067P: sources/stacks-chow/tags.txt, source lines 7978–7978. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 13517 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FF0: sources/stacks-chow/pinned-original-source.tex, source lines 13412–13412. Same-chapter pointer into the author environment lemma at source line 13412; invoked at proof line 13518 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 0F9Z: sources/stacks-chow/pinned-original-source.tex, source lines 6184–6184. Same-chapter pointer into the author environment remark at source line 6184; invoked at proof line 13521 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FBK: sources/stacks-chow/pinned-original-source.tex, source lines 11912–11912. Same-chapter pointer into the author environment lemma at source line 11912; invoked at proof line 13522 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FF1: sources/stacks-chow/pinned-original-source.tex, source lines 13435–13435. Same-chapter pointer into the author environment lemma at source line 13435; invoked at proof line 13526 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

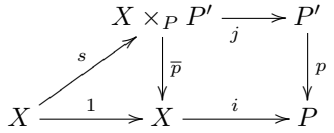
Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let $f : X \to Y$ be a local complete intersection morphism of schemes locally of finite type over S. The bivariant class $f^!$ is independent of the choice of the factorization $f = g \circ i$ with g smooth (provided one exists).*

Proof. Given a second such factorization $f = g' \circ i'$ we can consider the smooth morphism $g'' : P \times_Y P' \to Y$, the immersion $i'' : X \to P \times_Y P'$ and the factorization $f = g'' \circ i''$. Thus we may assume that we have a diagram

$$\begin{array}{ccccc} & & P' & & \\ & \overset{i'}{\nearrow} & \downarrow{\scriptstyle p} & \overset{g'}{\searrow} & \\ X & \xrightarrow{i} & P & \xrightarrow{g} & Y \end{array}$$

where p is a smooth morphism. Then $(g')^* = p^* \circ g^*$ (Lemma 02RD) and hence it suffices to show that $i^! = (i')^! \circ p^*$ in $A^*(X \to P)$. Consider the commutative

diagram



where $s = (1, i')$. Then s and j are regular immersions (by Divisors, Lemma 067R and Divisors, Lemma 067P) and $i' = j \circ s$. By Lemma 0FF0 we have $(i')^! = s^! \circ j^!$. Since the square is cartesian, the bivariant class $j^!$ is the restriction (Remark 0F9Z) of $i^!$ to P', see Lemma 0FBK. Since bivariant classes commute with flat pullbacks we find $j^! \circ p^* = \overline{p}^* \circ i^!$. Thus it suffices to show that $s^! \circ \overline{p}^* = \mathrm{id}$ which is done in Lemma 0FF1. $\square$